4 The graph of a wave at time $t = 0$ is shown in Fig. 4.1. The wave is travelling to the left with a speed of 340 m s^{-1}.

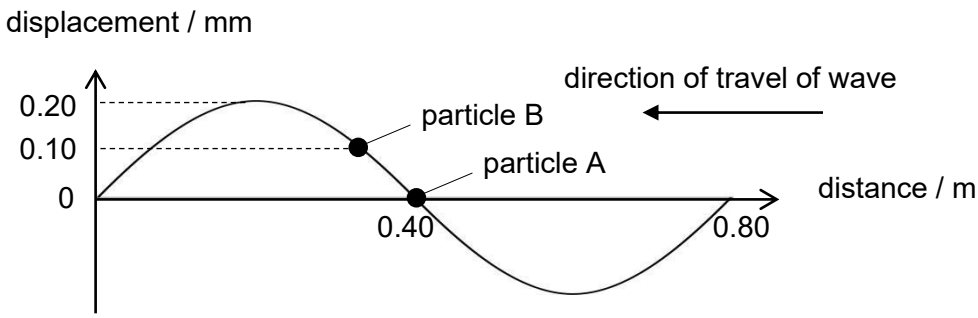

Fig. 4.1

(a) Calculate the frequency of the oscillation of particle A.

frequency = ... Hz [2]

(b) Show that the total energy E_T of the oscillation of particle A is given by

$$E_T = 2\pi^2 m f^2 a^2,$$

where a is the amplitude of vibration of particle A and m is the mass of particle A.

[2]

(c) The mass of particle A is 4.7×10^{-26} kg. Use the expression in **(ii)** to calculate the energy of the oscillations of particle A.

energy = ……. J [2]

(d) Sketch on the axes of Fig. 4.2 the variation of displacement with time for particle A.

Fig. 4.2

[2]

(e) Calculate the phase angle between the two particles A and B.

phase angle = …….° [2]

5 **(a)** State the *principle of superposition*